

Lesson plan

Name of the Programme:	B.Tech (Hons)	
Title of the Course:	Exploring Science	
Course code	CS1804	
Semester:	1	
Names of the Course Instructor(s)	Dr. Durgarao Guttula Dr. Subir Roy	
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
	2+0+2	30+30 = 60

Session	Module No.	Topic	Pedagogy / Activities	Pre-reading / Reference
1	1	Classification of solids based on band theory	Lecture, PPT, Discussion	Donald A. Neamen
2		Intrinsic and extrinsic semiconductors: Concept of electrons, holes, mobility	Lecture, Numerical Problems	Donald A. Neamen
3		Diode and its characteristics	Lecture, Demonstration	Donald A. Neamen
4		Zener diode and its characteristics	Lecture, Case Study	Donald A. Neamen
5		Transistor: Input & Output characteristics	Lecture, PPT	Donald A. Neamen
6		Optoelectronic devices: Photodiode, LED, Solar cells	Video, Discussion	Donald A. Neamen
7	2	Properties of crystals and crystal lattices	Lecture, Diagrams	B.D. Cullity
8		Unit cells, lattice parameters, and crystal systems	Lecture, 3D Models	B.D. Cullity
9		Bravais lattices: Primitive, BCC, FCC, Base-centered	Lecture, Models	B.D. Cullity
10		Miller indices: Concepts and calculations	Lecture, Problem Solving	B.D. Cullity
11		Atomic packing fraction: SC, BCC, FCC	Lecture, Numerical Problems	B.D. Cullity

12		Bragg's law and X-ray diffraction	Lecture, Video, Analysis	B.D. Cullity
13	3	Introduction to nanomaterials and miniaturization	Lecture, Discussion	Alain Nouailhat
14		Surface area to volume ratio, properties of nanomaterials	Lecture, Examples	Alain Nouailhat
15		Synthesis techniques: Top-down and bottom-up	Lecture, Videos	Alain Nouailhat
16		Synthesis: Photolithography, Ball milling	Lecture, Demo Videos	Alain Nouailhat
17		Chemical vapor deposition technique	Lecture, Case Studies	Alain Nouailhat
18		Applications of nanotechnology, nanoscience vs nanotechnology	Seminar, Group Discussion	Alain Nouailhat
19	4	Pre-quantum experiments (Young's, Blackbody, Hertz, etc.)	Lecture, Experiment Analysis	P.C. Deshmukh
20		Planck's law, Heisenberg's microscope, photoelectric effect	Lecture, Concept Discussion	P.C. Deshmukh
21		Quantum principles: Superposition, measurement, observables	Lecture, Simulation	P.C. Deshmukh
22		Hermitian operators and quantum states	Lecture, Problem Solving	P.C. Deshmukh
23		Schrödinger equation and solvable quantum systems	Lecture, Numerical Problems	P.C. Deshmukh
24		Introduction to quantum computing: Qubits, gates, entanglement	Video, Discussion	P.C. Deshmukh
25	5	Climate science and environmental modeling	Lecture, Case Study	Arthur Beiser
26		Environmental pollutants: Types and effects	Lecture, Discussion	Arthur Beiser
27		E-waste: Sources, composition, and health hazards	Lecture, Report Analysis	Arthur Beiser
28		E-waste management and recycling methods	Lecture, Case Study	Arthur Beiser
29		Water management: Techniques and sustainability	Lecture, Discussion	Arthur Beiser

30		Sustainable waste management strategies	Lecture, Report Analysis	Arthur Beiser
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Assessment and Evaluation

Continuous Internal Evaluation (CIE): 70 Marks	
Components of CIE	Weightage of each component
Multiple Choice Quiz	10 %
Presentation	10 %
Pen and Paper	25 %
Practical	25 %

Semester End Exams (SEE)	
Mode of Exam	Theory (Pen and Paper)
Weightage of exam	30 %

Course Outcomes – Programme Outcomes Matrix

CO No.	Course Outcome	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO 1	Explain semiconductor behavior, diode characteristics, and transistor operation based on band theory and charge carrier dynamics.	3	2	2	1	0	2	2	2	0	0	0	0



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CO 2	Analyze crystal structures using Miller indices and Bragg's law, and interpret X-ray diffraction patterns to determine material properties.	3	3	2	1	0	0	1	2	0	0	0	0
CO 3	Describe nanomaterial properties, synthesis methods, and their advantages in miniaturization.	3	3	3	2	0	0	1	2	0	0	0	0
CO 4	Evaluate environmental pollutants, e-waste hazards, and strategies for sustainable water and waste management.	2	3	2	1	1	1	2	2	1	2	2	3

Rubrics for evaluation of CIE (separate for each CIE component)

CIE-1 – MCQ

Programme Outcome	PO1, PO2, PO3, PO4, PO6, PO7, PO8
Total Marks	10
Quiz	MCQ only. Each correct answer carries 1 mark. No negative marking for wrong answers

CIE-1 – Presentation

Programme Outcome	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12					
Total Marks	20 (Scale down 10 marks)					
Criteria	CO-PO	Excellent	Very Good	Good	Average	Needs Improvement
Clarity (2 Marks)	PO1, PO2, PO3	The topic is clearly defined, depicting salient points and scope concisely and accurately. 2 Marks	The topic is mostly clear, covering most key points accurately. 1.5 Marks	The topic is somewhat clear, but some key points are missing or inaccurately presented. 1.Marks	The topic lacks clarity and is not well-defined. 0.5 Mark	The topic is poorly defined, and key points are missing or unclear. 0 Marks
Scientific Content (4 Marks)	PO1, PO2, PO3, PO4	Content is accurate with clear evidence supporting the topic. Presentation is specific and concise.	Content is mostly accurate, with some evidence provided. Presentation is generally specific but lacks detail.	Content is somewhat accurate, but lacks adequate evidence. Presentation is too general. 2 Marks	Content is inaccurate or lacks evidence, and presentation is vague. 1 Marks	Content is inaccurate, lacking evidence, and the presentation is unclear. 0.5 Marks

		4 Marks	3 Marks			
Presentatio n Quality (2 Marks)	PO5 , PO6 , PO7	Slides are easy to follow with clear captions/labels. Visuals are high-quality, and sources are cited. 2 Marks	Slides are mostly clear, with adequate visuals. Most sources are cited correctly. 1.5 Marks	Slides are somewhat difficult to follow, with low-quality visuals. Inconsistent source citation. 1 Mark	Slides are unclear with poor visuals and missing citations. 0.5 Mark	Slides are very unclear, visuals are poor, and citations are missing. 0 Marks
Knowledge (4 Marks)	PO1 , PO2 , PO3 , PO6	Demonstrates strong understanding, defends points effectively, and articulates clearly to the audience. 4 Marks	Demonstrates good understanding, defends most points well, and communicates adequately. 3 Marks	Demonstrates basic understanding, defends some points, but lacks clear communication. 2 Mark	Lacks understanding, fails to defend points, and struggles to communicate effectively. 1 Marks	Shows limited or no understanding and struggles with clear communication. 0.5 Marks
Delivery (2 Marks)	PO7 , PO8 , PO10	Voice is clear, pacing is well-managed, and technology is used appropriately. Finishes on time. 2 Marks	Voice and pacing are mostly clear, and technology is used effectively. Mostly finishes on time. 1.5 Marks	Voice is somewhat clear, pacing is inconsistent, and technology use is occasionally distracting. 1.Mark	Voice is unclear, pacing is poor, and technology is overused or misused. Does not finish on time. 0.5 Mark	Voice is very unclear, pacing is poor, and technology is misused or distracting. 0. Mark

Viva (6 Marks)	PO4 , PO6 , PO1 2	Answers questions with confidence and depth. 6-5 Marks	Answers questions adequately with good understanding. 4-3 Marks	Answers questions, but lacks depth and confidence. 3-2 Marks	Struggles to answer questions and shows limited understanding. 2-1 Marks	Struggles significantly to answer questions and shows little to no understanding. 0.5 Mark
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CIE-2: Midterm Examination

Programme Outcome	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9
Total Marks	25
CIE-2	Pen and paper. Evaluation will be done according to the Scheme of evaluation provided by the course faculty

CIE-3

Lab examination marks: 30

Lab record and behaviour marks: 20

Total: 50 (Scaled down to 25 marks)

Lab examination: 30 marks

Programme Outcome	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12					
Total Marks	30					
Sl. No.	Criteria	Measuring Method	Excellent	Good	Satisfactory	Poor
Write Up (5 Marks)						
1	Clarity and Organization	Written	Well-organized, clear, and concise with no	Clear and organized with minor spelling/gra	Generally clear but with some spelling/gra	Poorly organized and unclear with many

			spelling/grammar errors. (2 marks)	grammar errors. (2 marks)	grammar errors. (1 mark)	spelling/grammar errors. (0 marks)
2	Content	Written	Comprehensive coverage of the topic with in-depth explanation. (3 marks)	Good coverage with sufficient explanation. (3 marks)	Adequate coverage but lacks depth. (2 marks)	Very limited coverage and depth. (1–0 marks)
Data Collection (10 Marks)						
1	Quality of Data	Observation	Data is highly relevant and of excellent quality. (4 marks)	Data is relevant and of good quality. (3 marks)	Data is somewhat relevant and acceptable. (2–1 marks)	Data is irrelevant or poor quality. (0 marks)
2	Quantity of Data	Observation	Extensive data collected. (3 marks)	Sufficient data collected. (2 marks)	Adequate data collected. (1 mark)	Insufficient data collected. (0 marks)
3	Documentation	Observation	Data thoroughly documented. (3 marks)	Well documented. (2 marks)	Adequately documented. (1 mark)	Poorly documented. (0 marks)
Analysis/Graph (5 Marks)						
1	Accuracy and Insight	Observation	Analysis accurate and provides deep insights. (5 marks)	Mostly accurate and provides good insights. (4–3 marks)	Somewhat accurate, some insights. (2–1 marks)	Incorrect, no insights. (0 marks)

Result (2 Marks)						
1	Conclusion	Written	Results clearly stated, well-supported by data. (2 marks)	Clearly stated, mostly supported. (1 mark)	Somewhat clear, partially supported. (0.5 marks)	Missing or unsupported. (0 marks)
Viva (8 Marks)						
1	Understanding	Viva	Demonstrates thorough understanding. (5 marks)	Demonstrates good understanding. (4–3 marks)	Adequate understanding. (2–1 marks)	Limited understanding. (0 marks)
2	Response to questions	Viva	Responds accurately and confidently. (3 marks)	Responds accurately to most questions. (2 marks)	Responds adequately to some questions. (1 mark)	Unable to respond. (0 marks)

Lab Behaviour Assessment 20 marks

Sl. No	Criteria	Excellent (4)	Very Good (3)	Good (2)	Needs Improvement (0–1)
1	Attendance	100% punctual, never absent	Rarely absent, mostly on time	Occasional absences/lateness	Frequent absences, irregular
2	Discipline	Always follows lab rules and decorum	Minor lapses but generally disciplined	Needs reminders occasionally	Often careless, disruptive

3	Teamwork	Actively supports peers, shares tasks fairly	Cooperative and supportive most of the time	Works independently, less collaborative	Not cooperative, avoids teamwork
4	Punctuality	Always on time, well prepared	Rarely late, generally prepared	Sometimes late or unprepared	Often late, unprepared
5	Lab Safety & Conduct	Always follows safety norms, highly careful	Rare lapses, generally safe	Needs reminders for safety protocols	Unsafe practices, careless with equipment